WORKED SOLUTION

STEP I 2002, Question 4

Pure • Curve sketching

STEP I 2002, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q4

The key idea

A line through $(0,1)$ that touches the curve further along must lie *below* the curve between those two points, because the curve is concave there and then flattens out. A region under a straight line is a trapezium, and rotating it about the y-axis gives a cylinder and a cone, so its area and volume are easy to find without calculus. Comparing them with the exact area and volume under the curve gives the two inequalities.

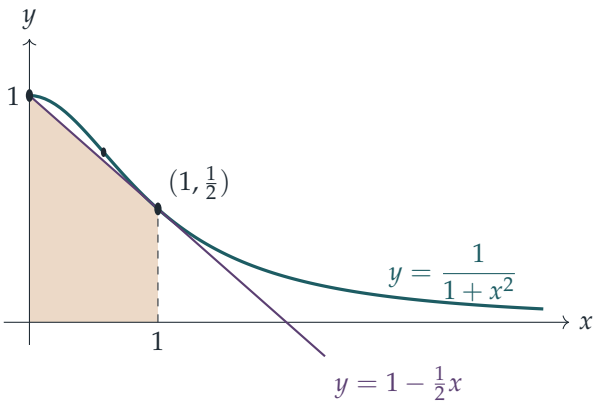
Solution

The sketch

Write $f(x) = \dfrac{1}{1+x^2}$. Then $f(0) = 1$, $f(x) > 0$, and $f(x) \to 0$ as $x \to \infty$, so the x-axis is an asymptote. Differentiating,

$$f'(x) = -\frac{2x}{(1+x^2)^2}, \qquad f''(x) = \frac{6x^2-2}{(1+x^2)^3}.$$

So $f'(0) = 0$ (the curve starts horizontally at $(0,1)$) and $f'(x) < 0$ for $x > 0$, so the curve decreases throughout. Also $f''(x) < 0$ for $0 \leq x < 1/\sqrt{3}$ and $f''(x) > 0$ for $x > 1/\sqrt{3}$, so there is a point of inflection at $(1/\sqrt{3}, \frac{3}{4})$: the curve is concave (bending down) near the y-axis and convex further out.



The shaded trapezium is the region under the line for $0 \le x \le 1$; it lies inside the region under the curve.

The line

The line passes through $(0, f(0)) = (0,1)$, so it is $y = 1 + mx$ for some gradient m. Suppose it is tangent at $x = a > 0$. Two conditions must hold there.

It meets the curve: $1 + ma = \dfrac{1}{1+a^2}$, so $ma = -\dfrac{a^2}{1+a^2}$, and since $a \neq 0$,

$$m = -\frac{a}{1+a^2}.$$

It has the curve's gradient: $m = f'(a) = -\dfrac{2a}{(1+a^2)^2}$.

Equating these and dividing by $-a/(1+a^2)$ (allowed, as $a > 0$) gives $1 = \dfrac{2}{1+a^2}$, so $a^2 = 1$ and, since $a > 0$, $a = 1$. Then $m = -\frac{1}{2}$, and the line is

$$y = 1 - \tfrac{1}{2}x,$$

touching the curve at $(1, \frac{1}{2})$.

No further intersections

Subtract the curve from the line:

$$\left(1 - \frac{x}{2}\right) - \frac{1}{1+x^2} = \frac{(2-x)(1+x^2) - 2}{2(1+x^2)} = \frac{-x^3 + 2x^2 - x}{2(1+x^2)} = -\frac{x(x-1)^2}{2(1+x^2)}.$$

This is zero only when $x = 0$ or $x = 1$. So the line meets the curve at $x = 0$ and at the point of tangency $x = 1$ (the repeated factor $(x-1)^2$ is the algebraic sign of tangency), and nowhere else: not for $x \ge 0$, and in fact not for negative x either.

We also read off something we need next. For $x > 0$ and $x \neq 1$ the expression is strictly negative, so

$$0 < 1 - \tfrac{1}{2}x < \frac{1}{1+x^2} \qquad \text{for } 0 < x < 1. \tag{$*$}$$

(The left inequality holds because $1 - \frac{1}{2}x \geq \frac{1}{2}$ on $[0,1]$.) So on $[0,1]$ the line lies strictly below the curve except at the endpoints, as in the sketch.

$\pi > 3$

Integrate $(*)$ over $[0,1]$. The two functions are continuous and the curve is strictly above the line on the open interval, so the inequality survives integration as a strict one:

$$\int_0^1 \frac{1}{1+x^2}\,\mathrm{d}x > \int_0^1 \Big(1 - \frac{x}{2}\Big)\mathrm{d}x.$$

The left side is $\frac{\pi}{4}$ (it is $[\arctan x]_0^1$). The right side is the area of the shaded trapezium, with parallel sides 1 and $\frac{1}{2}$ and width 1, so it is $\frac{1}{2}(1+\frac{1}{2}) = \frac{3}{4}$. Hence $\frac{\pi}{4} > \frac{3}{4}$, that is,

$$\pi > 3.$$

$\ln 2 > 2/3$

Now rotate the same two regions (for $0 \leq x \leq 1$) through 2π about the y-axis. By $(*)$, the trapezium sits inside the region under the curve, so its solid of revolution sits inside the other one, and has smaller volume.

> **Ask yourself** How do we find a volume of revolution about the y-axis when the curve is given as y in terms of x?

Use cylindrical shells: the thin strip at x of width δx and height y sweeps out a shell of volume about $2\pi x\, y\, \delta x$, so the volume is $\int_0^1 2\pi x\, y\,\mathrm{d}x$. (Alternatively, slice horizontally; see the Checks.)

For the curve,

$$V_{\text{curve}} = \int_0^1 \frac{2\pi x}{1+x^2}\,\mathrm{d}x = \pi[\ln(1+x^2)]_0^1 = \pi \ln 2,$$

since the numerator $2x$ is the derivative of the denominator.

For the line, the solid is a cylinder of radius 1 and height $\frac{1}{2}$ (from the part of the trapezium below $y = \frac{1}{2}$) topped by a cone of radius 1 and height $\frac{1}{2}$ (from the triangle above it). So

$$V_{\text{line}} = \pi \cdot 1^2 \cdot \tfrac{1}{2} + \tfrac{1}{3}\pi \cdot 1^2 \cdot \tfrac{1}{2} = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}.$$

As a check by shells: $\int_0^1 2\pi x(1 - \frac{1}{2}x)\mathrm{d}x = 2\pi(\frac{1}{2} - \frac{1}{6}) = \frac{2\pi}{3}$.

The integrand $2\pi x(\frac{1}{1+x^2} - 1 + \frac{x}{2})$ is continuous, non-negative on $[0,1]$ and strictly positive on $(0,1)$ by $(*)$, so $V_{\text{curve}} > V_{\text{line}}$:

$$\pi \ln 2 > \frac{2\pi}{3} \quad\Longrightarrow\quad \ln 2 > \frac{2}{3},$$

as required.

Checks

- Numerically, $\pi/4 \approx 0.7854 > 0.75$ and $\ln 2 \approx 0.6931 > 0.6667$, so both inequalities are true and neither is very slack.
- Slicing horizontally instead: for $\frac{1}{2} \le y \le 1$ the curve has $x^2 = \frac{1}{y} - 1$, so the volume is the cylinder $\pi \cdot 1^2 \cdot \frac{1}{2}$ plus $\int_{1/2}^{1} \pi\left(\frac{1}{y} - 1\right)\mathrm{d}y = \pi\left(\ln 2 - \frac{1}{2}\right)$, giving $\pi \ln 2$ again.
- Quick test of the factorisation: at $x = 2$ the line gives 0 and the curve $\frac{1}{5}$, a difference of $-\frac{1}{5}$; the formula gives $-\frac{2\cdot 1}{2\cdot 5} = -\frac{1}{5}$.

Common slips

- Finding the tangent at a general point and forgetting the condition that it passes through $(0, 1)$ (or vice versa): you need both conditions to pin down a.
- Stopping at "$\ge$": the strict inequalities need the line to be strictly below the curve on an interval, which is what the factor $x(x-1)^2$ gives you.
- Rotating about the x-axis by mistake, which gives $\int \pi y^2 \,\mathrm{d}x$ and different numbers.

Beyond STEP

The argument is a comparison principle: if $g \le h$ on an interval, with strict inequality somewhere and both continuous, then $\int g < \int h$. Making "strictly somewhere" precise is a typical exercise in a first analysis course, where the integral is defined via upper and lower sums. Bounding π by areas of simple shapes goes back to Archimedes, and much sharper versions of this question (such as $\int_0^1 x^4(1-x)^4/(1+x^2)\,\mathrm{d}x = \frac{22}{7} - \pi$) are worth looking up. For the analysis, try Abbott, *Understanding Analysis*.

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